Lab 05 – MATH 240 – Computational Statistics

Andrew Li
Colgate University
Department of Mathematics
ali@colgate.edu

Abstract

This experiment seek to analyze how big of an influence did different music groups have on songs, specifically how much did the bands Manchester Orchestra, The Front Bottoms, and All Get Out affect the song "Allentown." By using Essentia (Bogdanov et al., 2013), an open-source program for music analysis and synthesis, the musical components of 180 songs were isolated to be compared with the parts of "Allentown." This method would be give more depth into the analysis and provide a better understanding of what elements are extracted from music. Through this process, the steps of data cleaning and summarization will be explored.

Keywords: Cleaning data, extracting and automating data processing

1 Introduction

Music plays such an important role in everyday lives, from creating social spaces, providing an outlet, and connecting people throughout the world. As a result, the works of different artists can be traced to find influence from older generations or their peers. The song "Allentown" by Front Bottoms and Manchester Orchestra has influence from the two bands and All Get Out, so this lab seeks to explore what elements are similar and who overall has contributed the most to the creation of this song. There are many ways to conduct this comparison, but musical components was an objective way to analyze in pieces how similar the songs are. To do this, Essentia (Bogdanov et al., 2013) is going to be used to collect data via music analysis. Since Essentia allows processing on individual tracks, it became neccesary to prepare a batch of json files using the tracks from all the artists.

2 Methods

To replicate this experiment, it is important to make sure to download the jsonlite and stringr `R` packages. The music directory contained all 181 of the song tracks from the three bands, and we created a program that extracted all the provided .wav files (musical files) and processed them to create the batch file of executable command lines for the Essentia program to run. Using the `stringr` (Wickham, 2023) package for R , we were able to extract information from each wave file and organize each line by artist and album. Following that, we used the `jsonlite` (Ooms, 2014) package to create a sample program that can pull information from the json files that Essentia would return. This preliminary step showed how the larger scale data extraction will work.

To get a full picture on which band plays the most influence, we took the data from the Essentia processed calls, drew additional features such as mood from Essentia models (Alonso-Jiménez et al., 2020), and analyzed thoughts and personality traits through a text analysis tool called LIWC. This was all compiled into one `csv` file to sort and organize all of our data before we initiated a feature-by-feature comparison to see what insights we can draw about the rise of the song "Allentown."

Using boxplots, a sample analysis was shown for the emotional element for "Allentown" compared to the tracks from the three bands.

Using the tidyverse package for `R` (Wickham et al., 2019), it became easier to summerize and visualize our data because of the extensive functionality built into the package. Comparing individual values would have taken a large amount of space, lots of time, and difficulty to read, so to get a full analysis, we took all of the musical features, compared if the values of each feature from "Allentown" are similar to those from each artist, and graphed the proportion of songs that have similar traits in Figure 3. Using the `xtable` package (Dahl et al., 2019), a sample of some of the analytical work is shown in Table 1 before. The total mark of 1.00 would have meant that musical features of the artist's songs were not identical, but all very similar in style.

3 Results

Initially, it waas hard to determine what band had a greater influence on the song "Allentown." By collecting analyzed musical elements and processed components, it becomes possible to see what parts of the song are similar to the works of Manchester Orchestra, All Get Out, or The Front Bottoms. The final product from the data collection and cleaning is presented inside the trainingdata.csv file while the musical components for "Allentown" is separated inside the testingdata.csv to avoid conflicting overlap.

When looking at just the emotional aspect of the songs by the three bands in Figure 1, the dashed line represents the value of the emotional aspect of "Allentown," and comparing just this value would point to Manchester Orchestra as the main contributor since the song is within its IQR while being in the lower quartile for All Get Out and The Front Bottoms. However, when comparing the chords scale in Figure 2, more songs from The Front Bottoms share similar chords as "Allentown," so instead, Figure 3 showed the comparison for all traits . It can be determined that Manchester Orchestra has played the most influence on the creation of Allentown because it has a noticeable amount of similar traits in their music. While the songs were not exactly the same (1.00 being the maximum similarity), Manchester Orchestra's works is 10 percent greater than those of All Get Out and The Front Bottoms.

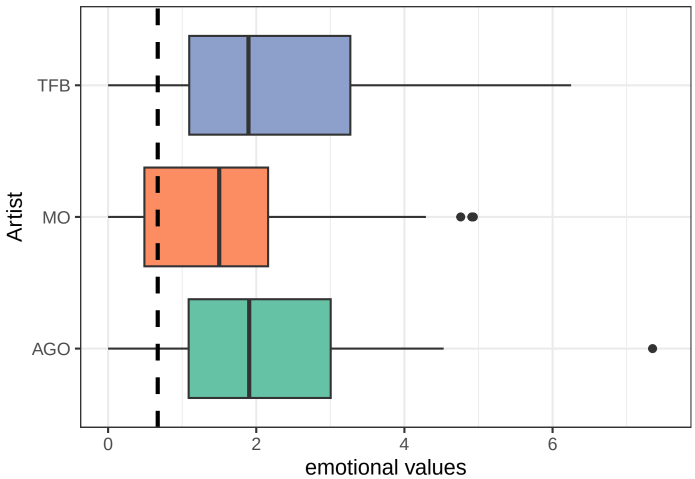

Figure 1: Emotional sound analysis
(names are shortened to MO, AGO, and TFB)

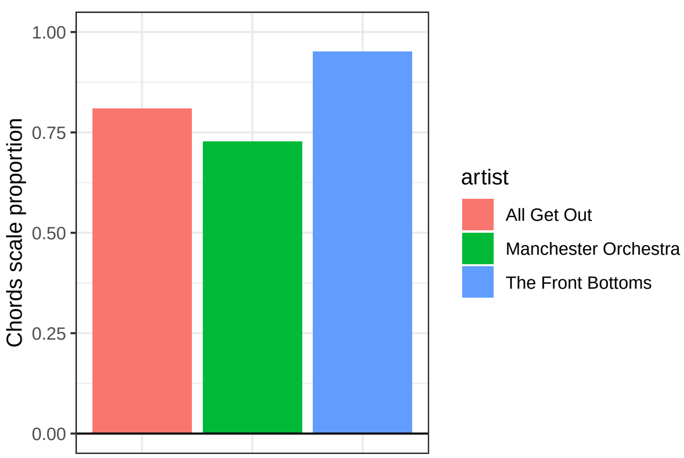

Figure 2: Songs similarity analysis based on chords scale

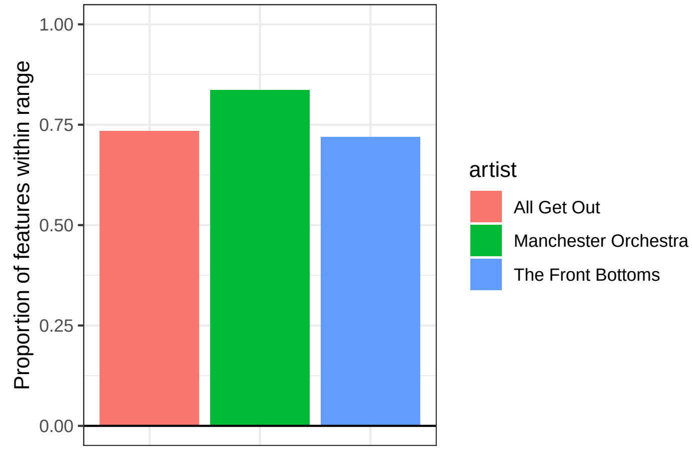

Figure 3: Songs similarity analysis based on total features

4 Discussion

Given Figure 1 and Figure 2, it can be seen how individual musical feature comparison gives an insufficient amount of information to determine which band had the greatest influence on the song "Allentown." After extracting wav files and creating additional analyzing additional features from them through Essentia models and LIWC, the final cummulative processing of data cleaning formed Figure 3 which shows how Manchester Orchestra contribued the most to the creation of the song. This type of analysis can be used to further determine the influence of other musicians on the works on any of these three bands or attribute similarities between different songs.

References

Alonso-Jiménez, P., Bogdanov, D., Pons, J., and Serra, X. (2020). Tensorflow audio models in essentia. In *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 266–270. IEEE.

Bogdanov, D., Wack, N., Gómez Gutiérrez, E., Gulati, S., Boyer, H., Mayor, O., Roma Trepat, G., Salamon, J., Zapata González, J. R., Serra, X., et al. (2013). Essentia: An audio analysis library for music information retrieval. In *Britto A, Gouyon F, Dixon S, editors. 14th Conference of the International Society for Music Information Retrieval (ISMIR); 2013 Nov 4-8; Curitiba, Brazil.[place unknown]: ISMIR; 2013. p. 493-8.* International Society for Music Information Retrieval (ISMIR).

Dahl, D. B., Scott, D., Roosen, C., Magnusson, A., and Swinton, J. (2019). *xtable: Export Tables to LaTeX or HTML*. R package version 1.8-4.

Ooms, J. (2014). The jsonlite package: A practical and consistent mapping between json data and r objects. *arXiv:1403.2805 [stat.CO]*.

Wickham, H. (2023). *stringr: Simple, Consistent Wrappers for Common String Operations*. R package version 1.5.1.

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., and Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43):1686.

5 Appendix

	artist	minimum	LF	RF	maximum	out.of.range	unusual	description
1	All Get Out	-14.27	-16.35	-5.10	-6.13	FALSE	FALSE	Within Range
2	Manchester Orchestra	-24.34	-20.22	-5.66	-6.26	FALSE	FALSE	Within Range
3	The Front Bottoms	-11.03	-9.65	-6.81	-5.71	FALSE	TRUE	Unusual
4	All Get Out	0.02	0.01	0.04	0.05	FALSE	FALSE	Within Range
5	Manchester Orchestra	0.01	-0.00	0.04	0.05	FALSE	FALSE	Within Range
6	The Front Bottoms	0.02	0.02	0.05	0.05	FALSE	FALSE	Within Range
7	All Get Out	0.84	0.77	1.18	1.30	FALSE	FALSE	Within Range
8	Manchester Orchestra	0.78	0.65	1.21	1.40	FALSE	FALSE	Within Range
9	The Front Bottoms	0.91	0.86	1.24	1.33	TRUE	FALSE	Out of Range
10	All Get Out	84.91	95.29	149.76	162.92	FALSE	FALSE	Within Range
11	Manchester Orchestra	67.15	71.18	160.30	184.57	FALSE	FALSE	Within Range
12	The Front Bottoms	80.83	82.99	157.05	165.67	FALSE	FALSE	Within Range
13	All Get Out	163.50	134.57	261.03	377.87	FALSE	FALSE	Within Range
14	Manchester Orchestra	102.43	79.15	320.67	434.36	FALSE	FALSE	Within Range
15	The Front Bottoms	108.49	127.14	258.48	297.35	FALSE	FALSE	Within Range
16	All Get Out	0.04	0.02	0.08	0.08	FALSE	FALSE	Within Range
17	Manchester Orchestra	0.02	0.00	0.07	0.09	FALSE	FALSE	Within Range
18	The Front Bottoms	0.04	0.04	0.07	0.11	TRUE	TRUE	Out of Range
19	All Get Out	0.43	0.26	1.05	1.29	FALSE	FALSE	Within Range
20	Manchester Orchestra	0.42	0.11	1.07	1.35	FALSE	FALSE	Within Range
21	The Front Bottoms	0.52	0.38	0.99	1.36	FALSE	FALSE	Within Range
22	All Get Out	2620015.25	3162397.09	4788882.53	4758639.50	TRUE	TRUE	Out of Range
23	Manchester Orchestra	1937052.25	1668651.00	4817223.00	5845988.50	FALSE	TRUE	Unusual
24	The Front Bottoms	3510343.75	3538674.50	5508843.50	5888673.00	FALSE	FALSE	Within Range

Table 1: Sample data from musical analysis